



C.1 Simple harmonic motion

Wave behaviour · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Simple harmonic motion (SHM) occurs when the restoring acceleration is directly proportional to displacement from equilibrium and always directed towards it: $a = -\omega^2 x$, where $\omega = 2\pi f$ is the angular frequency. Solutions have displacement $x = x_0 \sin(\omega t)$ or $x_0 \cos(\omega t)$ depending on initial conditions, velocity $v = \pm \omega \sqrt{(x_0^2 - x^2)}$ with maximum speed $v_{\text{max}} = \omega x_0$ at $x = 0$, and maximum acceleration $a_{\text{max}} = \omega^2 x_0$ at maximum displacement. Total mechanical energy in SHM (kinetic + elastic/gravitational potential) remains constant and equals $\frac{1}{2} m \omega^2 x_0^2$, continuously exchanging between kinetic and potential forms as the oscillator moves.

Worked example: follow the reasoning

A mass on a spring oscillates in SHM with amplitude 0.050 m and frequency 2.0 Hz. Find its maximum speed and maximum acceleration.

$$\omega = 2\pi f = 2\pi(2.0) = 12.57 \text{ rad}\cdot\text{s}^{-1}.$$

$$v_{\text{max}} = \omega x_0 = 12.57 \times 0.050 = 0.628 \text{ m}\cdot\text{s}^{-1}.$$

$$a_{\text{max}} = \omega^2 x_0 = (12.57)^2 \times 0.050 = 7.90 \text{ m}\cdot\text{s}^{-2}.$$

Answer: $v_{\text{max}} \approx 0.63 \text{ m}\cdot\text{s}^{-1}$, $a_{\text{max}} \approx 7.9 \text{ m}\cdot\text{s}^{-2}$.

Quick check · multiple choice

An object in SHM is at its maximum displacement from equilibrium. What are its velocity and acceleration at this instant?

- A Velocity zero, acceleration zero
- B Velocity zero, acceleration maximum
- C Velocity maximum, acceleration zero
- D Velocity maximum, acceleration maximum

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A 0.30 kg mass attached to a horizontal spring oscillates in SHM with amplitude 0.12 m and period 0.80 s. [8]
- (a) Calculate the angular frequency ω of the oscillation.
 - (b) Calculate the maximum speed of the mass.
 - (c) Calculate the total mechanical energy of the oscillation.
 - (d) Calculate the speed of the mass when its displacement is 0.060 m from equilibrium.

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $\omega = 2\pi/T = 2\pi/0.80 = 7.854 \text{ rad}\cdot\text{s}^{-1}$.

(b) $v_{\text{max}} = \omega x_0 = 7.854 \times 0.12 = 0.942 \text{ m}\cdot\text{s}^{-1}$.

(c) $E = \frac{1}{2}mv_{\text{max}}^2 = \frac{1}{2}(0.30)(0.942)^2 = 0.133 \text{ J}$.

(d) $v = \omega\sqrt{(x_0^2 - x^2)} = 7.854\sqrt{(0.12^2 - 0.060^2)} = 7.854\sqrt{(0.0144 - 0.0036)} = 7.854\sqrt{0.0108} = 7.854(0.1039) = 0.816 \text{ m}\cdot\text{s}^{-1}$.

Quick check answer

B. At maximum displacement the oscillator momentarily stops ($v = 0$) but the restoring acceleration is maximum since $a = -\omega^2 x$ is largest in magnitude at maximum $|x|$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.